UTM UNIVERSITI TEKNOLOGI MALAYSIA RESEARCH UNIVERSITY	**FACULTY OF COMPUTING** **UNIVERSITI TEKNOLOGI MALAYSIA**

WT (SECJ3483-03)
PROJECT PROPOSAL

Session/Semester: 20252026/ Semester 02

SECTION A: STUDENT INFORMATION

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SECTION C: PROJECT DETAILS

Project Title:	AI-Driven Incident Reporting System (AIRS)
Proposal No.*	1
Project Idea:	My own idea
Project Type:	System Development

Problem Background:

DHL Customer Support teams face significant delays and inconsistencies due to high volumes of incident reports received through fragmented channels like WhatsApp, Teams, and emails. Because this data is unstructured (screenshots, handwritten notes, and messy text), it is difficult to identify the core issue, assign it to the right department, or track resolution progress efficiently. This leads to slow response times and repetitive manual work.

Proposed Solution:

The **AI-Driven Incident Reporting System (AIRS)** is an automated solution that bridges raw data collection with structured management.

- **Data Ingestion:** UiPath bots fetch raw data from disparate sources.
- **Intelligence:** An LLM processes the "messy" data into clean, categorized summaries.
- **Management:** A centralized MERN-based web dashboard allows staff to track, assign, and resolve incidents in a standardized workflow.

Objectives:

- To automate the collection of incident reports from emails and Google Drive.
- To standardize unstructured input into clear, actionable summaries using AI.
- To centralize incident tracking through a secured web interface for better management oversight.
- To improve response times by automatically classifying and detecting duplicate incidents.

Scopes:

1. Functional Requirements (The "What")

- User Authentication: A secured login system for DHL employees.
- Automated Data Fetching: UiPath integration to pull data from internal DHL sources.
- AI Processing: Automatic summarization of long messages and extraction of key problems.
- Incident Dashboard: A workflow-based view where incidents can be tracked (Pending → Assigned → Resolved).
- Search & Filter: Ability to manage and search through historical incident records.
- Duplicate Detection: AI-powered identification of repeating issues.

2. Non-Functional Requirements (The "How")

- Security: Use of JWT and encryption to protect sensitive DHL logistics data.
- Scalability: Built on the MERN stack to handle increasing report volumes.
- Usability: A clean, intuitive React-based interface for fast navigation.
- Reliability: Error reporting and summary logs for the automation process.

Project Requirements:

Software	• **Operating System:** Windows 11 • **Development Environment: Visual Studio Code** for MERN stack development and **UiPath Studio** for automation workflows. • **Version Control: GitHub Desktop** for managing code iterations and collaboration within the team. • **Database Management: MongoDB Compass** for visualizing incident data collections. • **Design Tools: Figma** for prototyping the dashboard UI and incident status workflows.
Hardware	• **Processor:** Minimum Intel Core i5 or AMD Ryzen 5 • **Memory:** 16GB RAM • **Storage:** 512GB SSD

	• **Devices:** Mobile device for testing cross-platform responsiveness of the web app (React-based).
Technology/Technique / Method/Algorithm	• **MERN Stack:** Primary technology framework (MongoDB, Express, React, Node.js) for full-stack delivery. • **RPA (Robotic Process Automation):** Utilizing **UiPath** for automated data ingestion from messy sources like Google Drive and Outlook. • **Natural Language Processing (NLP):** Using **Large Language Models (LLM)** to implement text summarization and intent extraction algorithms. • **OCR (Optical Character Recognition):** Technique to extract text from images or screenshots of damaged packages. • **Duplicate Detection Algorithm:** A similarity-checking algorithm (such as Cosine Similarity) to identify repeating incident reports.
Network Elements	• **RESTful API:** To enable seamless communication between the **UiPath** bot (client) and the **Node.js** server (backend). • **Cloud Hosting:** Integration with services like **MongoDB Atlas** for cloud-based database accessibility. • **HTTPS Protocol:** Ensuring all data transmitted between the frontend dashboard and backend server is encrypted.
Security Elements	• **Authentication & Authorization:** Implementation of **JWT (JSON Web Tokens)** to secure the "Secured Website" login. • **Input Sanitization:** To prevent SQL/NoSQL injection when handling "messy" raw content from external sources. • **Role-Based Access Control (RBAC):** Restricting incident status updates to specific DHL authorized personnel only. • **API Key Management:** Securing LLM and UiPath Orchestrator API keys using environment variables (.env).
Project Area:	• **Primary Area: Digital Automation & AI-Driven Resource Allocation.**

	• **Secondary Area: Full-Stack Web Development & Logistics Process Optimization.** • **Context:** Specifically designed for the **DHL APSSC** Logistics Operations environment in **April 2026**.